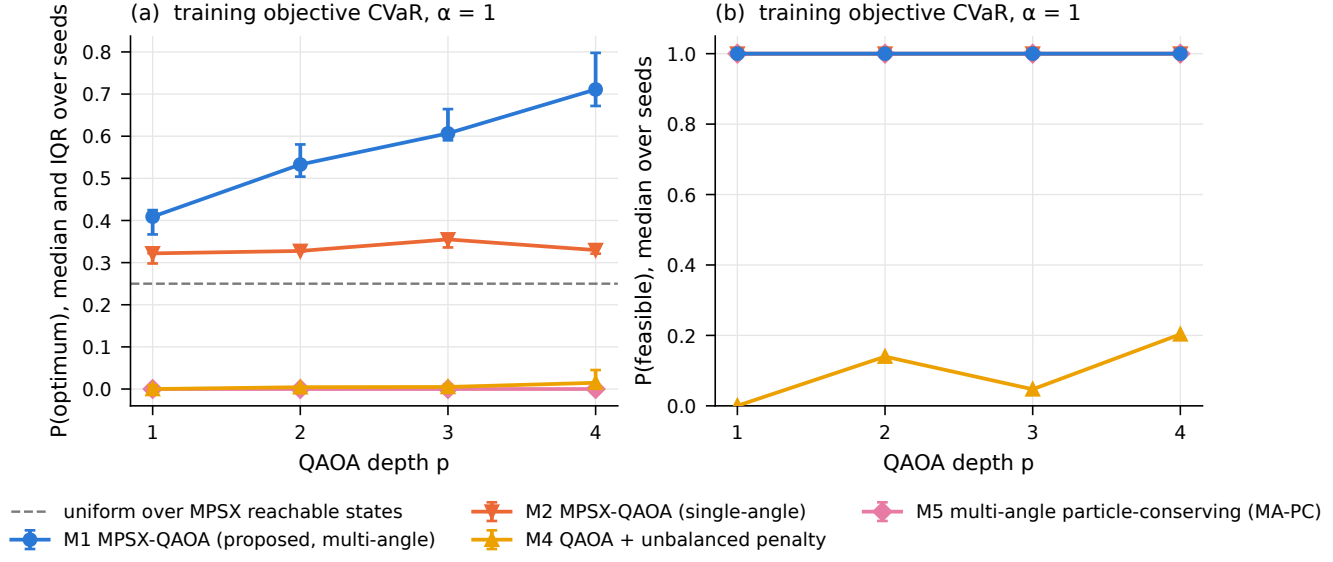
(a) training objective CVaR, α = 1
P(optimum), median and IQR over seeds
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
1
2
3
4
QAOA depth p
(b) training objective CVaR, α = 1
P(feasible), median over seeds
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
QAOA depth p
uniform over MPSX reachable states
M1 MPSX-QAOA (proposed, multi-angle)
M2 MPSX-QAOA (single-angle)
M4 QAOA + unbalanced penalty
M5 multi-angle particle-conserving (MA-PC)